AHB_Bus_Matrix USF Documentation

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1 AHB_Bus_Matrix Introduction

The AHB_Bus_Matrix_3M_6S models a three-manager, six-subordinate matrix with 32-bit address and data paths. Each manager can address every subordinate, and transfers to different subordinates can proceed concurrently. Each subordinate has independent fixed-priority arbitration: manager_2 has priority over manager_1, which has priority over manager_0. Arbitration recognizes only HTRANS=2'b10 (NONSEQ). When a subordinate is not ready, its arbiter retains the previous data-phase manager selection instead of granting a new request. There is no round-robin fairness or starvation prevention. This implementation is restricted to single transfers: IDLE is inactive, and BUSY/SEQ encodings do not request a grant. It has no HBURST or lock interface and does not implement burst sequencing or locked transfers. The one-bit HRESP supports OKAY and ERROR, not RETRY or SPLIT; full AMBA AHB compliance is not implied. The fixed, inclusive address windows in the XML match the RTL decoder: subordinate_0 (RTL SlaveInterface_A), 0x11FFF800-0x1203FFFF; subordinate_1 (B), 0x18000000-0x18007FFF; subordinate_2 (C), 0x00000000-0x00007FFF; subordinate_3 (D), 0x40000000-0x47FFFFFF; subordinate_4 (E), 0x48000000-0x5FFFFFFF; subordinate_5 (F), 0x17FFE000-0x17FFFFFF. Addresses are forwarded unchanged, without translation. An accepted NONSEQ access outside all windows produces a local two-cycle ERROR response: HRESP=1 with HREADY=0, followed by HRESP=1 with HREADY=1. Responses from mapped subordinates are routed back to the corresponding manager. No address-alignment, transfer-size, or protection-permission checks are implemented. HSIZE is forwarded unchanged; on the 32-bit data path, protocol-valid transfer sizes are byte, halfword, and word, not wider transfers. HMANAGER (named HMASTER in the RTL) is a four-bit, manager-supplied sideband value that is cached and forwarded unchanged; it is neither generated from the port number nor used for routing or priority. HPROT is likewise forwarded without enforcing access permissions or implementing caching or buffering. Each manager input stage caches HTRANS, HADDR, HMANAGER, HPROT, HWRITE, and HSIZE. In the RTL, these registers capture the live inputs on rising clock edges while that manager's HREADY is high and retain them while it is low; the routing path uses live inputs when ready and cached values while stalled. These are internal registers, not software-accessible storage. The RTL uses a common rising-edge clock and active-high asynchronous reset. Write and read data are routed combinationally using data-phase selections; the matrix does not buffer HWDATA or HRDATA. Wait states can extend the data phase. A subordinate's HREADYIN is the raw ready signal of the previous subordinate selected by its currently selected manager, or 1 when there is no applicable selection; it is not a command to insert wait states and does not include the manager's arbitration/error-ready gating. The XML describes behavior through request, response, inter-subordinate HREADYIN, and unmapped-address error traces, rather than one global implementation FSM. Its two-bit manager_N_subordinate_select_S helpers hold per-pair transfer-type values, not subordinate indices; manager selection uses 0 for no manager and 1/2/3 for manager_0/1/2. The delayed selection supports data-phase ownership, not just stalls. Model scope limits remain: the istage_N_valid_stall_S qualifiers are referenced but have no defining actions in this XML, and the inter-subordinate traces specify selected HREADYIN-high cases rather than its complete RTL behavior. These descriptions do not establish full RTL/USF equivalence.

2 AHB_Bus_Matrix Block Diagram

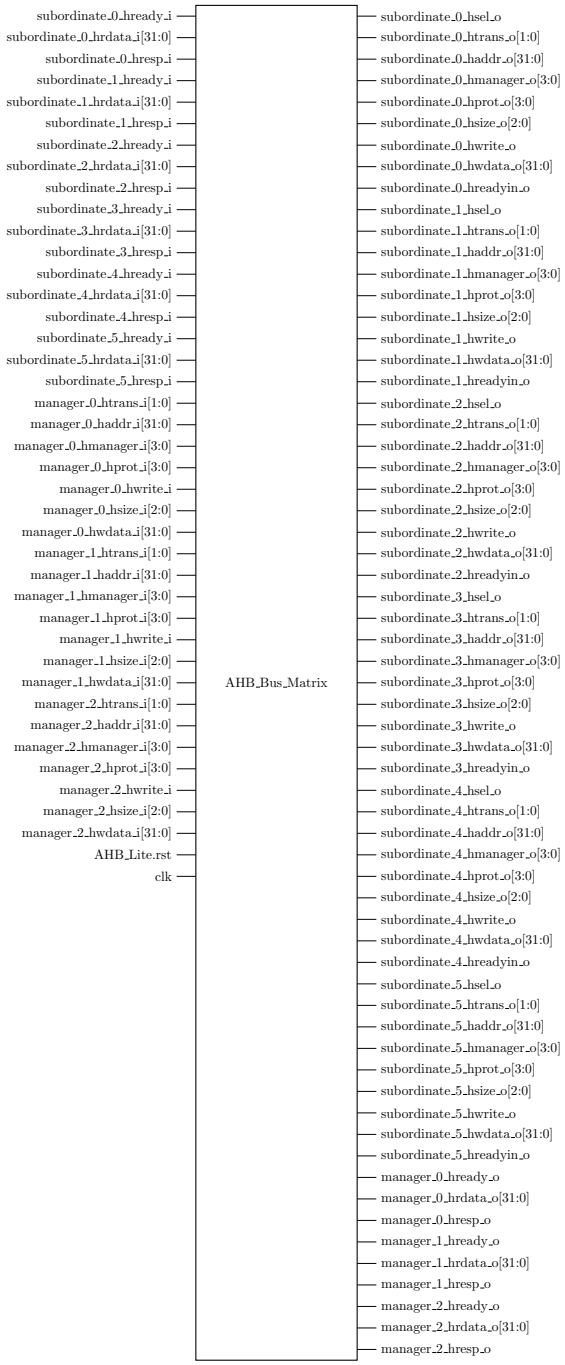


Figure 1: Top-level block diagram with primary inputs (left) and outputs (right).

3 AHB_Bus_Matrix Primary Inputs

Input	Description
subordinate_0_hready_i	The subordinate_0_hready_i input is the one-bit, active-high ready response from subordinate 0. A high value allows the current data phase to complete; a low value inserts a wait state, which ends when the signal is asserted high again. The RTL uses this input both to retain or update subordinate 0's arbitration selection and to route completion status to the data-phase manager. A manager's returned HREADY can also be held low by arbitration or local error handling, so it is not always identical to this input.
subordinate_0_hrdata_i	The subordinate_0_hrdata_i input carries 32-bit read data from subordinate 0. The RTL routes it combinationally to the data-phase manager using the delayed selections; there is no read-data capture register in the matrix. The connected manager samples the result when its read transfer completes with HREADY high. The XML response traces specify the returned read data on the immediate-response and delayed-response completion paths.
subordinate_0_hresp_i	The subordinate_0_hresp_i input is the one-bit response from subordinate 0: 0 means OKAY and 1 means ERROR. It is routed to the data-phase manager, with the RTL manager response gated by a registered NONSEQ indication. HRESP must be interpreted together with HREADY: an ERROR indication can be present before completion while HREADY is low. The matrix does not generate a two-cycle sequence for errors returned by a mapped subordinate; the subordinate supplies its own response timing. RETRY and SPLIT responses are not supported by this interface.

manager_0_htrans_i	The manager_0_htrans_i input carries the two-bit transfer type. Although AHB defines IDLE=00, BUSY=01, NONSEQ=10, and SEQ=11, this RTL's arbitration and active-transfer error detection test specifically for NONSEQ. The XML request and unmapped-address traces likewise require HTRANS=2 and manager_0_hready_o=1 to start a new transfer. BUSY and SEQ do not request arbitration and are not supported transfer modes. Some XML response bookkeeping copies any non-IDLE encoding, but that does not implement sequential or burst transfers. Use separate NONSEQ transfers for supported accesses; the input stage retains the transfer type while the manager is stalled.
manager_0_haddr_i	The manager_0_haddr_i input is manager 0's 32-bit address-phase address. The matrix decodes the live input when manager_0_hready_o is high and the retained input-stage address while it is low. The six fixed, inclusive address windows listed in the component description select subordinates 0 through 5; the selected subordinate receives the original address without translation. An accepted NONSEQ access outside every window generates a local ERROR response. No address-alignment check is implemented.
manager_0_hmanager_i	The manager_0_hmanager_i input is a four-bit sideband value named HMASTER in the RTL. The input stage caches it with the address/control signals and the routing multiplexers forward it unchanged to the selected subordinate. The matrix neither requires it to equal manager 0's port index nor checks its value. Address decoding uses HADDR, arbitration uses NONSEQ requests and fixed port priority, and response routing uses internal selection codes; none of these functions uses HMANAGER. Identifier meaning and uniqueness are system-level responsibilities.

manager_0_hprot_i	The manager_0_hprot_i input carries four-bit HPROT attributes. Conventionally, bit 0 distinguishes data from opcode access, bit 1 privileged from user access, bit 2 marks bufferable access, and bit 3 marks cacheable access. The matrix caches and forwards the bits unchanged with the address phase. It does not use them for decoding or arbitration, enforce protection permissions, or provide a cache or write buffer. Any such interpretation or enforcement must be supplied by the connected system.
manager_0_hwrite_i	The manager_0_hwrite_i input is a 1-bit wide write control signal received from manager interface 0 of the AHB Bus Matrix. When this input is HIGH, it indicates that the current transfer is a write operation. When this input is LOW, it indicates that the current transfer is a read operation.
manager_0_hsize_i	The manager_0_hsize_i input supplies the three-bit transfer-size encoding and is cached and forwarded unchanged. HSIZE=000, 001, and 010 specify byte, halfword, and word transfers respectively, the protocol-valid sizes for this 32-bit data path. Wider encodings do not enable wider transfers, bursting, or width conversion. The RTL does not validate HSIZE, check alignment, or repack data according to transfer size; the manager must provide a legal, appropriately aligned access supported by the subordinate.
manager_0_hwdata_i	The manager_0_hwdata_i input carries manager 0's 32-bit write data for the data phase of a write transfer. Unlike the address/control signals, HWDATA is not cached by the input stage: it passes combinationally through routing selected by delayed subordinate and manager selections. The manager must supply the data corresponding to the preceding write address phase and hold it through any wait states until completion. The matrix does not buffer, align, or resize the data.

Table 1: List of the AHB_Bus_Matrix's primary inputs.

4 AHB_Bus_Matrix Primary Outputs

Output	Description
subordinate_0_hsel_o	The subordinate_0_hsel_o output is the one-bit HSEL signal to subordinate 0. It is asserted when the selected manager's decoded address targets this subordinate and is low when no manager is selected. Selection is determined independently for each subordinate, so more than one subordinate can be selected concurrently by different managers. HSEL is an address-phase qualifier, not a transfer-completion indication; it must be interpreted with HTRANS and HREADYIN. Only NONSEQ requests participate in arbitration in this implementation.
subordinate_0_htrans_o	The subordinate_0_htrans_o output forwards the selected manager's two-bit transfer type, using its live or cached address-phase value, and defaults to IDLE (2'b00) when no manager is selected. The AHB encodings are IDLE=00, BUSY=01, NONSEQ=10, and SEQ=11, but this matrix recognizes only NONSEQ as an arbitration request. Forwarding a two-bit field does not provide BUSY, sequential-transfer, or burst support. The request traces likewise start only for HTRANS=2.
subordinate_0_haddr_o	The subordinate_0_haddr_o output is the 32-bit address-phase address of the manager selected for subordinate 0. It comes from that manager's live input or retained input-stage address and is forwarded without address translation. Subordinate 0 corresponds to RTL SlaveInterface_A and is decoded for the inclusive range 0x11FFF800-0x1203FFFF. In the RTL, the address output defaults to zero when no manager is selected; HSEL, HTRANS, and HREADYIN determine whether an address phase is valid.

subordinate_0_hmanager_o	The subordinate_0_hmanager_o output forwards the selected manager's four-bit HMANAGER value, named HMASTER in the RTL, from the live or cached address-phase signals. The matrix does not generate this value from the manager port number, validate it, or use it for address decoding or arbitration. Its meaning is supplied by the connected system. It is distinct from the internal two-bit manager-selection code. The RTL drives zero when no manager is selected.
subordinate_0_hprot_o	The subordinate_0_hprot_o output forwards the selected manager's four-bit HPROT address-phase attributes unchanged. The conventional meanings are data versus opcode access, privileged versus user access, bufferable, and cacheable. The matrix preserves these attributes through its input-stage cache and routing multiplexers, but does not check access permissions or implement buffering or caching based on them. Interpretation or enforcement belongs to the connected system. The RTL drives zero when no manager is selected.
subordinate_0_hsize_o	The subordinate_0_hsize_o output forwards the selected manager's three-bit HSIZE value unchanged. HSIZE encodes the number of bytes in one transfer as 2 to the power of HSIZE; it does not encode burst length. On this matrix's 32-bit data path, protocol-valid sizes are 000 for a byte, 001 for a halfword, and 010 for a word. Larger encodings can pass through the multiplexers but do not imply wider-transfer support or data-width conversion. The matrix neither validates HSIZE nor checks address alignment; managers and subordinates must obey the applicable size and alignment requirements.
subordinate_0_hwrite_o	The subordinate_0_hwrite_o output is a 1-bit wide write control signal that the AHB Bus Matrix forwards to the subordinate 0 interface. It indicates the transfer direction of the current AHB transaction, where a logic HIGH value specifies a write operation and a logic LOW value specifies a read operation. The Bus Matrix drives this output based on the HWRITE signal of the master that the arbitration logic has granted access to subordinate 0.

subordinate_0_hwdata_o	The subordinate_0_hwdata_o output is the 32-bit write-data path to subordinate 0. The RTL routes the manager's current HWDATA through the delayed subordinate and manager selections, so data-phase ownership can differ from the current address-phase grant. Write data is not registered, buffered, or reformatted by the matrix. The data phase follows the corresponding address phase and can extend across wait states; the manager must supply and hold the appropriate data until completion. The subordinate response traces explicitly forward this data during both waiting and completing data phases.
subordinate_0_hreadyin_o	The subordinate_0_hreadyin_o output supplies HREADYIN to subordinate 0. In the RTL, the current manager-selection code chooses a manager, and that manager's delayed address decode chooses the raw HREADY input of its previous subordinate. Consequently, this output can depend on another subordinate when a manager changes targets between transfers. It defaults to 1 when there is no selected manager or no valid delayed subordinate selection. It is not the selected manager's arbitration/error-gated HREADY output and is not a command for subordinate 0 to generate wait states. A low value indicates that the preceding data phase on the selected manager's path has not completed, so a new address phase must wait. The XML inter-subordinate traces constrain selected high-value cases only; they do not describe all low-value propagation or default cases implemented by the RTL.
manager_0_hready_o	The manager_0_hready_o output indicates whether manager 0 may advance: 1 allows the current data phase to complete and the next address phase to advance, while 0 extends the transfer. In the RTL it combines the delayed subordinate response path, registered arbitration-stall indications, and local unmapped-address error handling. A high value does not imply a successful transfer; HRESP distinguishes OKAY from ERROR. This signal also selects live versus cached address/control inputs and enables their cache updates. The XML marks HREADY as a register and assigns it in timed traces, with ResetValue=1; the RTL output itself is combinational logic driven partly by registered state, not a separate HREADY output register.

manager_0_hrdata_o	The manager_0_hrdata_o output is a 32-bit wide read-data signal that the AHB Bus Matrix returns to the manager interface upon completion of a read transfer. It carries the data retrieved from the addressed subordinate and presents it to the connected manager at the end of the applicable read transaction. The manager samples this output on the rising edge of the clock when the transfer completes.
manager_0_hresp_o	The manager_0_hresp_o output is the one-bit response to manager 0: 0 means OKAY and 1 means ERROR. For a mapped transfer it returns the data-phase subordinate response; for an accepted NONSEQ access outside all address windows the local error logic produces HRESP=1 with HREADY=0, followed by HRESP=1 with HREADY=1. In the RTL, HRESP is forced to zero when the registered input-stage NONSEQ indication is inactive. Completion requires HREADY=1, so HRESP alone does not indicate that the transfer has finished. No local alignment, size, or protection checks, and no RETRY or SPLIT responses, are implemented.

Table 2: List of the AHB_Bus_Matrix's primary outputs.

5 AHB_Bus_Matrix Registers

Register	Description
istage_0_htrans_r	The istage_0_htrans_r object models manager 0's internal two-bit HTRANS cache, with reset value zero. In the RTL it captures manager_0_htrans_i on a rising clock edge when manager_0_hready_o is high and retains its value while ready is low. Routing uses the retained value during the stall and returns to the live input when ready is high. The XML stall-entry and stall-retention actions express this using PAST values. Only a cached NONSEQ value participates as an active arbitration request; storing other encodings does not add support for them.
istage_0_haddr_r	The istage_0_haddr_r object models the 32-bit internal address cache for manager 0, not a software-accessible read/write register. Its reset value is zero. The RTL captures manager_0_haddr_i on rising clock edges while manager_0_hready_o is high, then retains and uses the cached address for decoding and forwarding while ready is low. When ready is high, the address path bypasses the cache and uses the live input. The XML captures the previous-cycle input on stall entry and preserves the previous-cycle cache value on continued stalls.
istage_0_hmanager_r	The istage_0_hmanager_r object models the four-bit HMANAGER sideband cache belonging to manager 0's input stage, corresponding to HMASTER in the RTL. It resets to zero, captures the manager-supplied value on rising clock edges while manager_0_hready_o is high, and retains it while ready is low. The cached value is forwarded during stalls; the live input is used when ready is high. This register preserves sideband metadata only: it does not determine transaction ownership, routing, or arbitration priority.

istage_0_hprot_r	The istage_0_hprot_r object models manager 0's four-bit internal HPROT cache, with reset value zero. The RTL captures the live protection attributes on rising clock edges while manager_0_hready_o is high and retains and forwards them while ready is low. Routing uses the live attributes again when ready is high. The register preserves the supplied bits without interpreting or enforcing protection, cacheability, or bufferability; the XML represents stall entry and retention with PAST assignments.
istage_0_hwrite_r	The istage_0_hwrite_r object models manager 0's one-bit internal HWRITE cache: 1 denotes write and 0 denotes read, with reset value zero. In the RTL it captures the live input on rising clock edges while manager_0_hready_o is high and retains it while ready is low. The retained direction is forwarded during stalls, and the live input is forwarded when ready is high. The XML models stall entry and retention with previous-cycle values. This is address-phase control storage, not write-data storage.
istage_0_hsize_r	The istage_0_hsize_r object models manager 0's three-bit internal HSIZE cache, with reset value zero. The RTL captures the live size encoding on rising clock edges while manager_0_hready_o is high, retains and forwards it while ready is low, and uses the live input when ready is high. The XML represents stall entry and retention with PAST assignments. The cached field is forwarded unchanged and is not used to select arbitration priority, validate transfer size, check alignment, or convert data width.

Table 3: List of the AHB_Bus_Matrix's registers.

6 AHB_Bus_Matrix Traces

6.1 manager_0_to_subordinate_0 Trace

The manager_0_to_subordinate_0 trace describes forwarding of a NONSEQ address/control request, not completion of its data phase. The zero-length request_transfer entry transition reaches transmit when manager_0_hready_o=1, manager_0_htrans_i=2, and the address lies in subordinate 0's inclusive window 0x11FFF800-0x1203FFFF. The zero-length bypass_matrix transition reaches out and forwards live address/control signals when subordinate 0 is ready and neither higher-priority manager 1 nor manager 2 has a live or cached request for it. Thus bypass_matrix is the immediately granted arbitration path, not a bypass of arbitration. Otherwise stall_transfer1 takes one cycle to stall_in_p; stall_transfer2 then enters stall, copies the previous-cycle address/control inputs into istage_0 registers, and sets manager_0_hready_o=0. Once arbitration permits forwarding, forward_stalled reaches out using the cached signals. If the request remains blocked and istage_0_valid_stall_0 is asserted, keep_stalling1 takes one cycle to stall_keep_p and keep_stalling2 returns to stall, retaining the cached values and low HREADY. That qualifier has no defining action in this XML. Both forwarding paths assert HSEL, forward HADDR/HTRANS/HMANAGER/HPROT/HWRITE/HSIZE, set the per-pair transfer helper to 2, and set the manager-selection code to 1. The final out state marks address/control forwarding only; separate response traces handle HWDATA, HRDATA, HRESP, and data-phase completion. It does not imply immediate release of data-phase ownership.

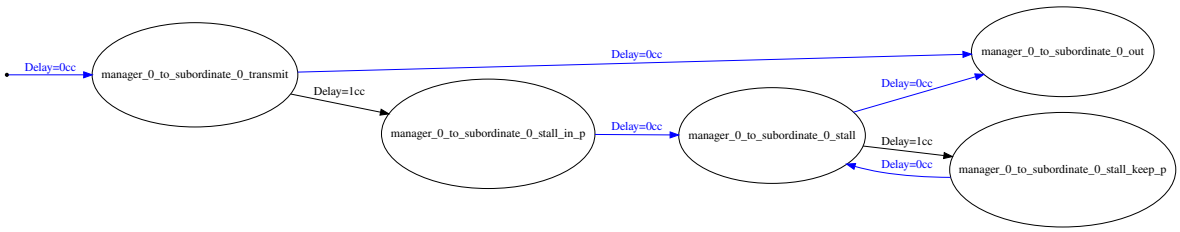


Figure 2: Trace visualization as state chart.

6.2 subordinate_0_to_manager_0 Trace

The subordinate_0_to_manager_0 trace describes the data phase following a forwarded manager 0 request to subordinate 0. Its subordinated_addressed entry transition has length one and reaches access after subordinate_0_hready_i=1, HSEL=1, the per-pair transfer helper equals 2, and the manager-selection code equals 1. Thus immediate_response means no additional subordinate wait state in the data phase, not zero latency from the original address phase. When subordinate 0 is ready, immediate_response reaches the final response state, sets manager_0_hready_o=1, forwards HRDATA and HRESP to manager 0, and forwards the current manager HWDATA to subordinate 0. When it is not ready, subordinate_stalls enters stall, sets manager HREADY low, retains the per-pair transfer helper, records the previous manager-selection code, and forwards HWDATA. The length-one keeps_stalling_p transition reaches stall_p while the subordinate remains not ready and the retained selection still identifies this transfer. From stall_p, keeps_stalling returns to stall with retained selection and low HREADY, or delayed_response reaches response when the subordinate is ready and performs the completion routing. Write data remains manager-supplied rather than buffered by these traces. Both completion paths also update the per-pair transfer helper from the live non-IDLE HTRANS and address, or clear it to zero; that bookkeeping does not add support for BUSY or SEQ requests.

6.3 s0_2_s1 Trace

The s0_2_s1 trace constrains HREADYIN at subordinate 1 in selected cases involving a manager previously associated with subordinate 0. It does not transfer data between subordinates. The zero-length trigger_trace transition enters access when any manager's subordinate-0 transfer helper equals 2 and

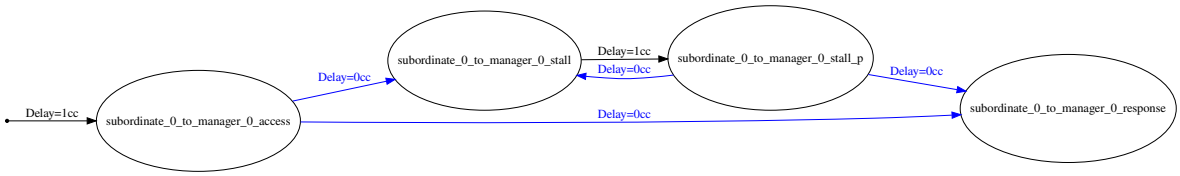


Figure 3: Trace visualization as state chart.

subordinate_0_manager_select_delayed identifies that manager. The length-one access_to_direct transition requires subordinate_0_hsel_o and reaches direct_response; direct_to_responded then requires both subordinate ready inputs high and subordinate_1_manager_select equal to the previous-cycle subordinate_0_manager_select. The length-one access_to_delayed transition instead uses the entry selection condition and reaches delayed_response; delayed_to_responded requires both ready inputs high and subordinate_1_manager_select equal to the previous-cycle subordinate_0_manager_select_delayed. Each completion transition assigns only subordinate_1_hreadyin_o=1 and reaches responded. The direct and delayed branch predicates are not mutually exclusive, and neither branch is selected simply by a ready-versus-not-ready test. There is no waiting loop or HREADYIN=0 action in this trace. It documents a subset of the RTL's ready routing when changing subordinate targets, not a general subordinate-to-subordinate transaction or the complete HREADYIN function.

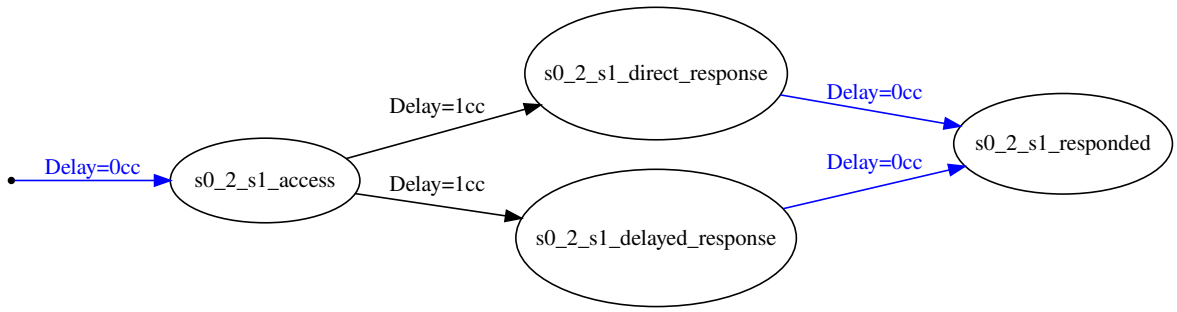


Figure 4: Trace visualization as state chart.

6.4 m0_error Trace

The m0_error trace specifies manager 0's local response to an unmapped NONSEQ access. The zero-length m0_error_trace_trace transition is the entry to access, not a transition spanning the whole trace: it requires manager_0_hready_o=1, manager_0_htrans_i=2, and an address outside every inclusive subordinate window. The length-one access_to_error transition has the same trigger and assigns manager_0_hresp_o=1 and manager_0_hready_o=0 for the first error-response cycle. The unconditional length-one error_to_resp transition then assigns HRESP=1 and HREADY=1 and reaches the final resp state, completing the transfer in the second error-response cycle. No real subordinate is selected for the unmapped request. This trace covers address-decode errors only, not alignment, size, or protection violations, and it is separate from forwarding errors returned by mapped subordinates. The corresponding RTL combines its unmapped-address check, registered NONSEQ indication, and two-state error-ready FSM.

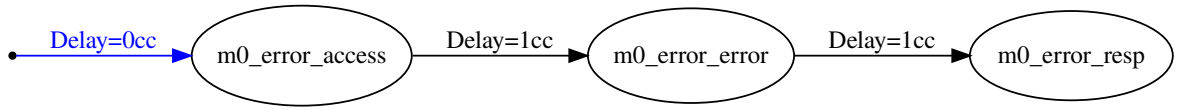


Figure 5: Trace visualization as state chart.